

COMBINED SCIENCE

FORM 1-2

BIOLOGY NOTES

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BIOLOGY SECTION

- **LABORATORY RULES AND SAFETY**
- **CELLS AND LEVELS OF ORGANISATION**
- **NUTRITION**
- **RESPIRATORY SYSTEMS**
- **TRANSPORT SYSTEM**
- **REPRODUCTION**
- **HEALTH AND DISEASE**

LABORATORY RULES AND SAFETY

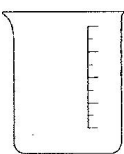
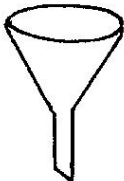
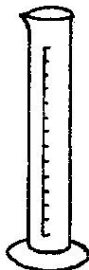
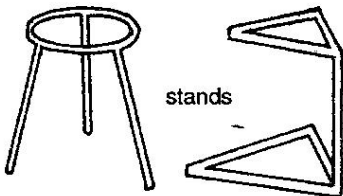
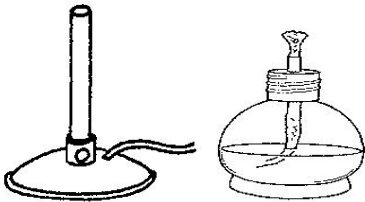
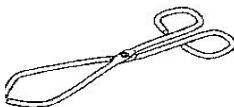
LABORATORY SAFETY AND APPARATUS

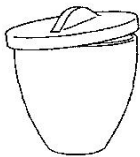
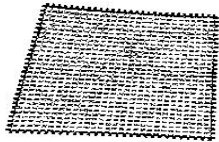
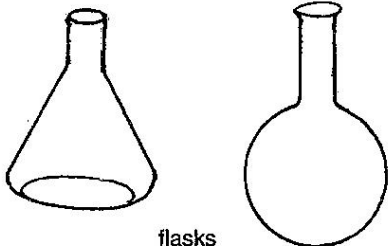
Laboratory rules

- A lab is a room equipped for scientific research, experimentation and analysis.
- Laboratory rules are a set of instructions that guide us in the laboratory.
- Safety in the laboratory is essential to prevent serious accidents to yourself and others, to avoid damage of apparatus, ensures accurate record keeping and to avoid injuries
- The following are a set of lab rules;
 - ✓ Do not work in the lab without your teacher
 - ✓ Read and follow instructions
 - ✓ Read labels on all chemical bottles before use
 - ✓ Do not use any chemical until the teacher has explained its uses and safety precaution to be taken
 - ✓ Do not perform any experiments and do not touch materials or apparatus without instructions from your teacher
 - ✓ Do not taste anything unless instructed to do so since some chemicals are toxic
 - ✓ Report all accidents to the teacher even if they appear to minor
 - ✓ In case of skin contact with chemicals, wash immediately with running water
 - ✓ Use burners with great care to avoid fires
 - ✓ Wash hands with soap after handling chemicals
 - ✓ Clean your work areas and equipments after use
 - ✓ Always wear safety clothing (lab coats, gloves & goggles)

Laboratory apparatus

- Apparatus refers to the set of equipments used by scientists to carry out a scientific investigation

Apparatus	Diagram	Function
Beaker		- used for heating, holding and measuring liquids though it is not very accurate.
Test tube		- Hold small amounts of chemicals or liquids
Funnel		- Channels liquid or powdery solids from one container into another or can be used with filter paper to separate substances.
Measuring cylinder		- Is used to measure out the volume of liquids.
Tripod stand		- Have 3 legs and is used to support various apparatus above the flame during heating.
Burners		- Provides a flame used for heating substances
Tongs		- used to hold hot apparatus
Spatula		- Is used to scoop small amounts of solid chemicals from containers.

Crucible		- It is ceramic containers capable of withstanding extreme temperatures. They hold solids and small volumes of liquids for heating.
Wire gauze		- Is placed under a container that is being heated so that the container does not have direct contact with the flame. It spreads the heat from the burner and prevents glass from breaking
Evaporating dish		- used for evaporating solutions with or without heating
Flasks	 flasks	- used for heating, holding and measuring liquids though it is not very accurate.

CELLS AND LEVELS OF ORGANISATION

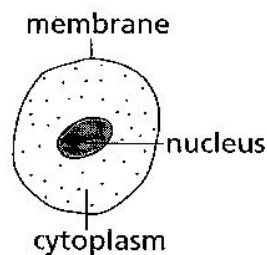
CELLS

PLANT AND ANIMAL CELLS

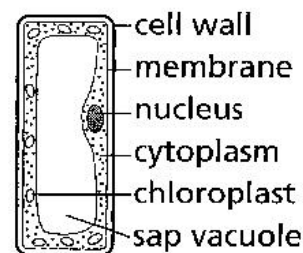
- A cell is the basic unit building block of all living things.
- It is the fundamental unit of structure and function of life. It may perform a specific function in the body of a living organism.
- Living things can be classified as multi-cellular i.e. are made of many cells or unicellular organisms that are composed of only a single cell.

Structure of a plant and animal

Animal cell



Plant cell



Cell membrane

It is a partially permeable layer that forms a boundary around the cytoplasm. It retains the cell contents and controls the movement of substances into and out of the cell. In this way the cell membrane maintains the structure and chemical reactions of the cytoplasm.

Cytoplasm

This is a jelly-like substance which contains particles and organelles. It is the site in which all chemical reactions take place in the cell.

Nucleus

It is always embedded in the cytoplasm. Control the type and quantity of enzymes produced by the cytoplasm. In this way it controls all the activities of the cell. It carries genetic information (DNA in form of chromosomes).

Cell wall

It is a tough non-living layer made of cellulose and other compounds surrounding the cell membranes. It confers shape and, to some extent, rigidity on the cells. It is permeable to water and other substances into and out of the cell. It prevents plant cells from bursting

Vacuole

It is a fluid filled space surrounded by a membrane. It contains cell sap containing salts and sugars dissolved in water. The outward pressure of the vacuole on cell wall makes the plant cells firm, giving strength and resilience to the tissues (keeps the cell rigid). Animal cells may sometimes have small vacuoles but are usually produced to do a particular job and are not permanent.

Chloroplasts

Contain the green pigment (chlorophyll) which absorbs light for photosynthesis.

Similarities and differences between plant and animal cells

Similarities

- All have cell membrane, cytoplasm and nucleus

Differences

- Plant cells are usually larger and their walls give them a distinct outline
- Plant cells are surrounded by a cell wall made of cellulose
- Often plant cells have chloroplasts.
- Plant cells usually have a thin lining of cytoplasm, with a large central vacuole.

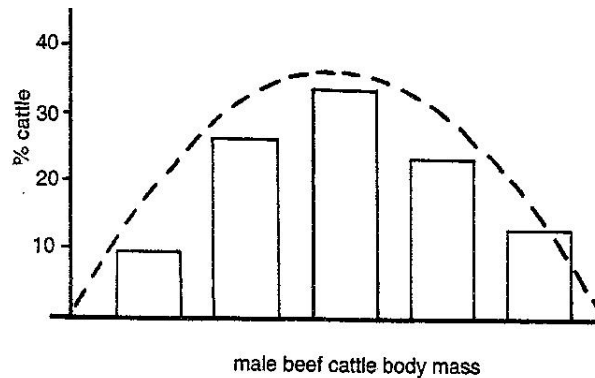
VARIATION

- variation is the observable difference among organisms with reference to specific characteristics.
- variations that can be inherited are determined by genes while other variations can be caused by the environment or a combination of both genes and the environment.
- Differences among organisms include height, sex, mass, earlobes, coat or skin colour, left or right handedness

Types of variation

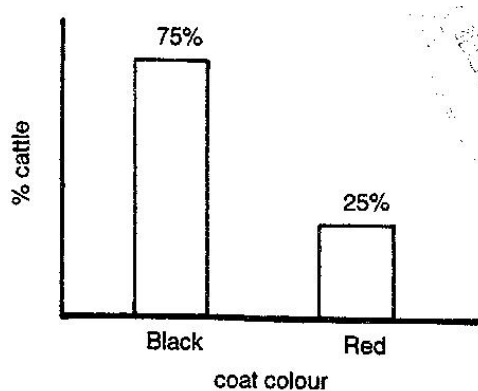
Continuous variation

- Show some intermediates and have no clear cut differences. Occurs where differences between organisms show a gradual change. It is due to interactions between the environment and the genotype is most likely to be influenced by the environment e.g. height, mass, seeds in a pod, shoe size.



Discontinuous variation

- Occurs where there are clear cut differences organisms with no intermediates. Most features of discontinuous variation are clearly genetically determined and there is a limited number of alternatives e.g. tongue rolling, presence or absence of earlobes in humans, sex, eye colour, albinism, blood groups, coat colour.

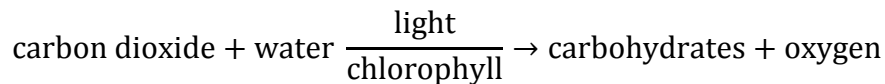


NUTRITION

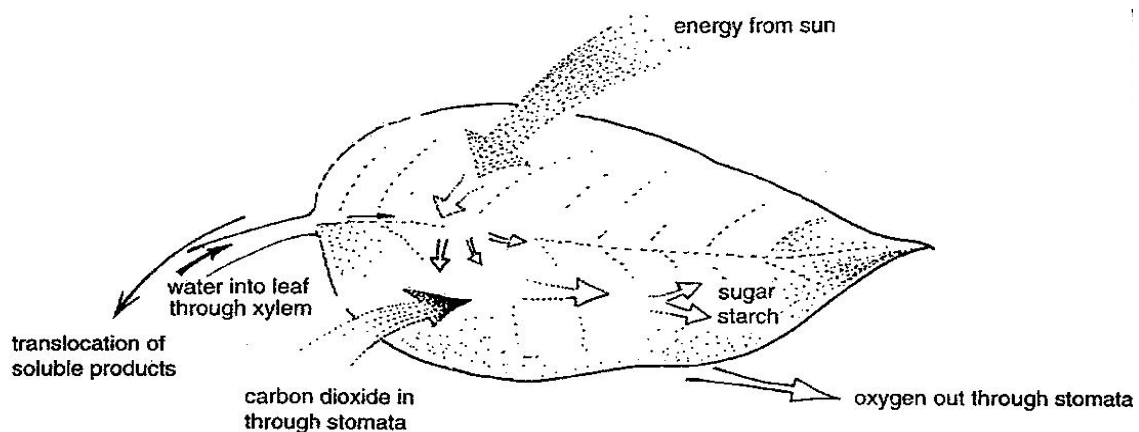
PLANT NUTRITION

PHOTOSYNTHESIS

- Is the production of carbohydrates by green plants using carbon dioxide and water in the presence of light energy



- Green plants convert light energy from the sun to chemical energy in carbohydrates during photosynthesis and later into other organic substances like starch.
- Plants also make loads of other nutrients for other organism to eat (consumers) e.g. amino acids, vitamins, fats which are stored in leaves, roots, fruits and seeds.
- They produce oxygen needed by organisms for respiration.
- Green plants are autotrophy i.e. they make their own organic molecules from inorganic components. They are producers
- Animals are heterotrophy i.e. they feed on other organisms or their products to fuel life processes. They are consumers.



- Carbon dioxide and water enters the leaf cells. Chlorophyll in the chloroplasts traps light energy from the sun. the light energy is used to split water into hydrogen and oxygen (photolysis)
- The oxygen escapes from the leaf to the atmosphere as a by-product. The hydrogen reacts with carbon dioxide to form carbohydrates.

Raw materials

1. Water is absorbed by root hairs and transported through the xylem
2. Carbon dioxide diffuses through the stomata

Conditions needed

1. Chlorophyll essential to absorb solar energy
2. Sunlight

End products

1. Carbohydrates can be used immediately by the plant or stored in the form of starch in seeds, fruits, stems and roots.
2. Oxygen diffuses out of the leaf or can be used for respiration by the plants and animals.

Importance of photosynthesis

- Produces food for all living organisms
- Produces oxygen which is used by plants and animals during respiration.
- Helps to reduce global warming by absorbing carbon dioxide
- Converts light energy into chemical energy

ANIMAL NUTRITION

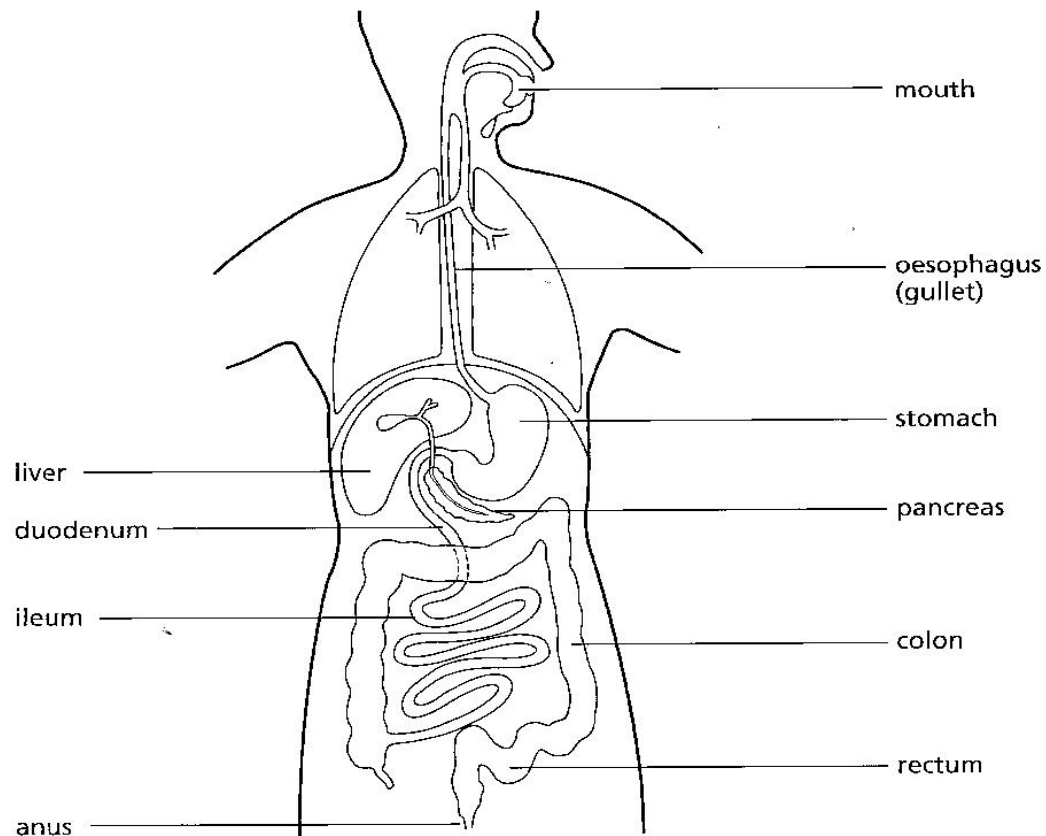
DIGESTIVE SYSTEM OF HUMANS

- The alimentary canal is a long tube which runs from the mouth to the anus. It is part of the digestive system which also includes liver, pancreas and gall bladder.

Parts of digestive system

- Food is taken into the digestive system through the mouth (ingestion). In the mouth the food is chemically digested by enzymes and mechanically digested by the teeth.
- When swallowed it passes through the esophagus and movement is facilitated by peristalsis.
- Food enters the stomach when it is churned for about 3-4 hours and mixed with digestive juices.
- Food enters the duodenum where there is chemical digestion of food by enzymes produced from the pancreas.

- In the ileum digestion is completed and nutrients absorbed into the blood stream by diffusion.
- Water and salts are absorbed in the colon by osmosis and active uptake.
- Waste matter is temporarily stored in the rectum and then passed out through the anus (Egestion).



BALANCED DIET

- It is a diet that includes all the essential nutrients in their correct proportions to keep one healthy

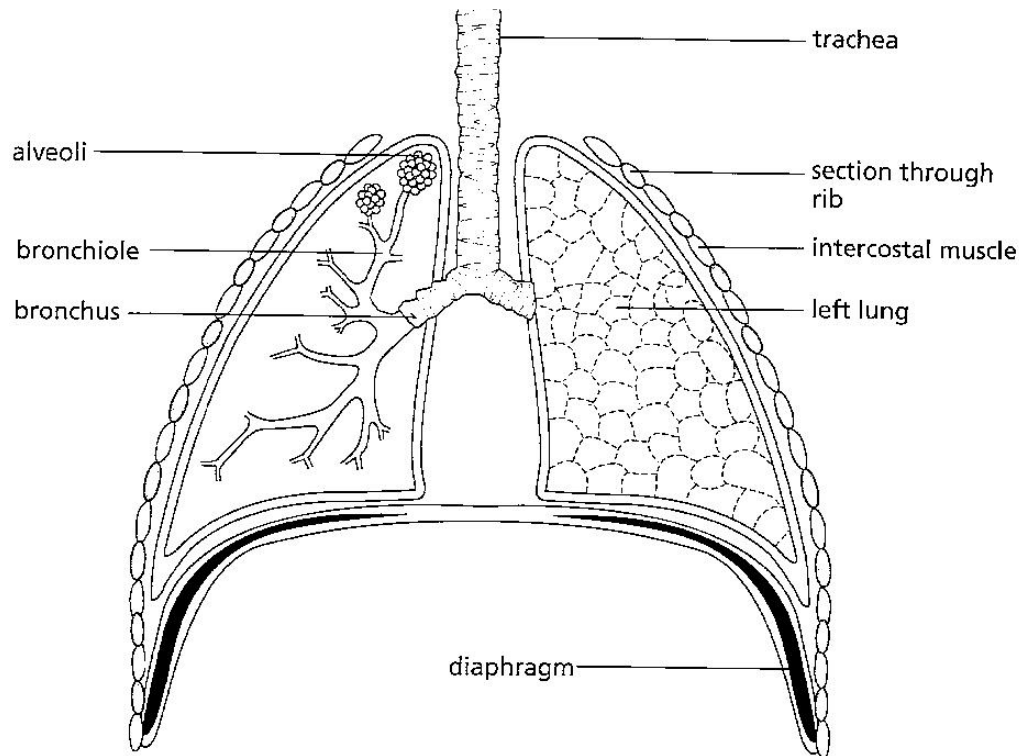
Components of a balanced diet

Component	Functions of nutrients	Food source
Carbohydrates	- Provide the body with energy	Bread, maize, yams, sweet potatoes, rice
Fats	- Provide energy - Store energy in the body - Reduce heat loss through the skin	Meat, butter, oil, milk, cheese, yoghurt, nuts

	- Protect organs in the body from physical damage	
Proteins	- Needed for growth, especially of muscles and nerves - Needed to repair damaged tissues	Meat, chicken, fish, eggs, beans, milk products
Vitamins	- Important for fighting infections - Improves wound healing - Prevents blindness	Fruits, vegetables, meat, milk, eggs
Mineral salts	- Needed for formation of strong bones and teeth - Needed for blood formation - Needed for growth and development	Vegetables, meat, milk, nuts
Fibre	- Assists digestion - Needed for normal bowel movement - Prevents constipation	Fruit, vegetables, wholegrain foods
Water	- Enables chemical reactions to take place in the body. - Transport substances around the body - Cools the body	Water melons

RESPIRATORY SYSTEM

RESPIRATORY ORGANS



- Air is taken through the nostrils and then travels via the trachea into the right and left bronchi, which divides and re-branches into bronchioles, each of which ends in a cluster of alveoli. It is just in the alveoli that the actual gaseous exchange occurs

Parts of respiratory system

Nose and nasal passages

- The nostrils contains hairs that assist in filtering out dirt from entering air
- The nasal passages warm the air and add moisture to the air. Mucus in the nasal passages traps bacteria and dust. Mucus is antiseptic i.e. kills bacteria.

Trachea (wind pipe), bronchi and bronchioles

- walls of the trachea are supported by c-shaped rings of cartilage. This prevents the trachea from collapsing and thus holds it open for free passage of air.
- The trachea divides into bronchi. Each bronchus extends to each lung and branches into numerous small bronchioles which further channels air to and from the alveoli.

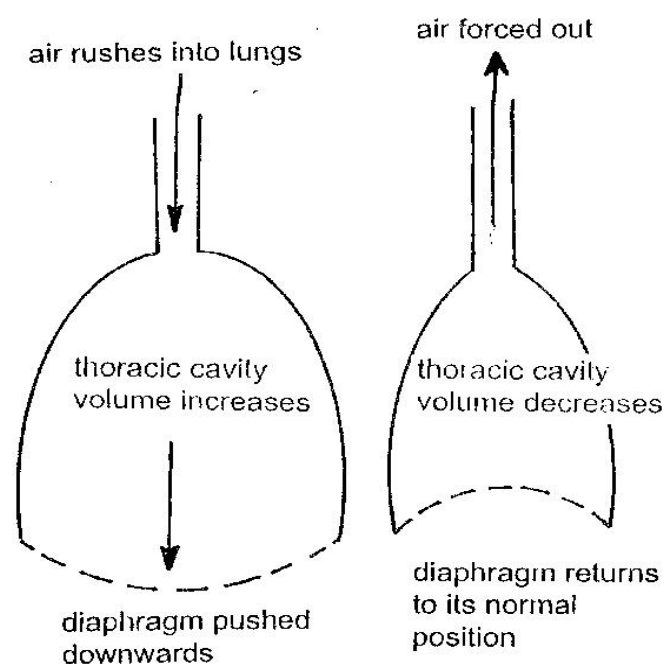
- Particles of dust and bacteria are trapped in the film of mucus covering these cells
- The trachea and its branches are also lined with cilia which are in constant motion and carry dust and dirt mixed with mucus, upwards, towards the mouth.

The air sacs

- The bronchioles branch repeatedly into smaller and smaller passageways which end in air sacs (alveoli)
- These are the respiratory surfaces where gaseous exchange takes place
- The air sacs are thin walled and covered by numerous blood capillaries to facilitate efficient gaseous exchange.

Breathing mechanism

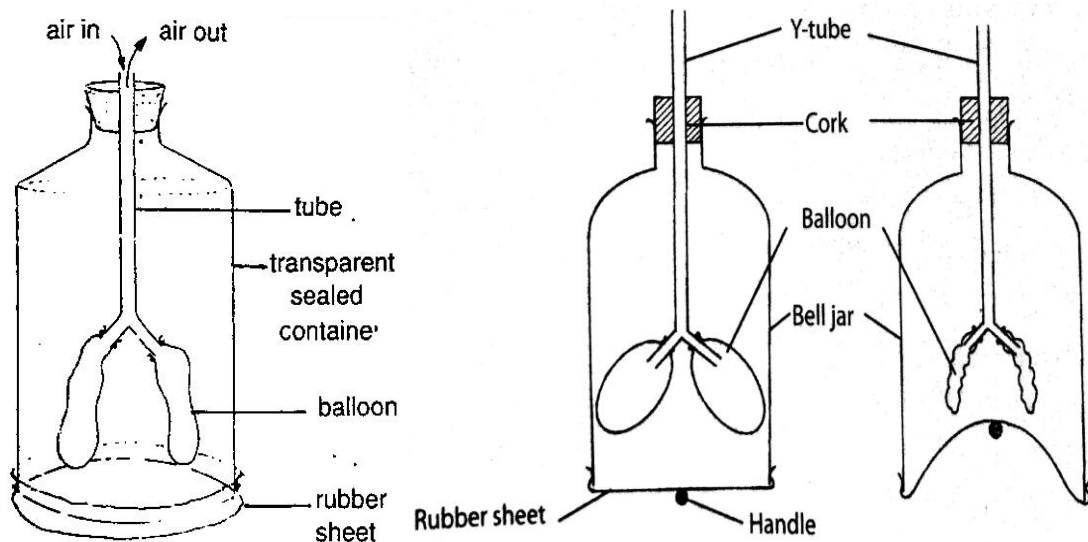
- Breathing is the ventilation of lungs by the movement of ribs and diaphragm causing air to enter lungs to bring oxygen and leave lungs to remove carbon dioxide
- Breathing involves two sets of muscles i.e. diaphragm and intercostal muscles which work to increase or decrease the volume of the thoracic cavity, so that pressure is increased or decreased and air moves in and out of lungs



- During breathing in (inhaling);
 - The external intercostal muscle contract while internal intercostal muscles relax, lifting the rib cage upwards and outwards. The diaphragm muscle contracts and moving the diaphragm down (flattens). The thoracic volume increases and pressure of air in the thoracic cavity increases. Air moves into the lungs.

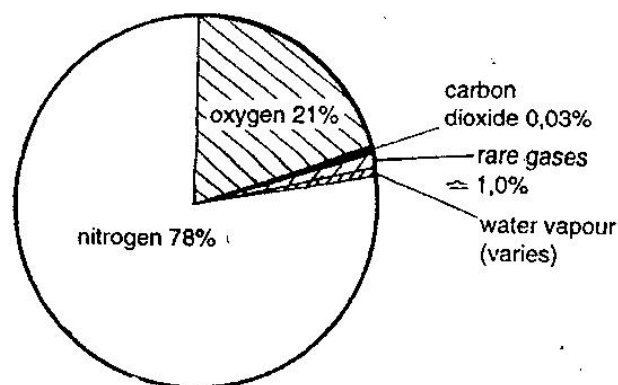
- During breathing out (exhaling);
 - The external intercostal muscles relax while internal intercostal contract, lowering the ribcage downwards and inwards. The diaphragm muscle relaxes and the diaphragm arches upwards. Thoracic volume decreases and pressure is increased. Air is forced out the lungs.

Demonstrating breathing



- A piece of apparatus called bell – jar model can be used to show the way in which movement of the diaphragm results in inspiration and expiration.
- The balloon starts off deflated. When the rubber sheet is pulled down, the balloons inflate, if released the balloon deflate again.
- When the rubber sheet is pulled down, the volume inside the bell jar increases. This reduces the air inside, making it lower than the outside. The air rushes in through the glass tubing, to equalise the air pressure, causing the balloons to inflate

The composition of air

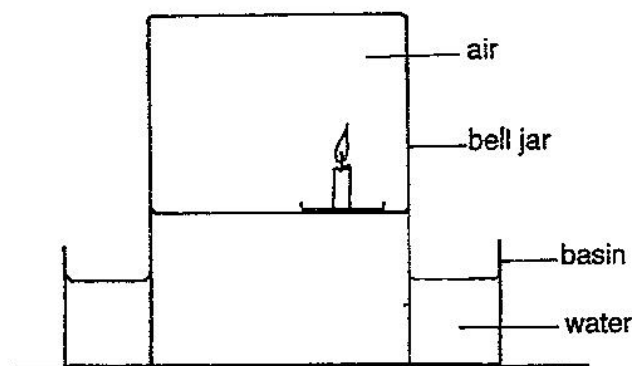


- Air is a mixture of gases i.e. nitrogen, oxygen, carbon dioxide, rare gases and water
- Air we breathe in contains about 21% Oxygen, some is used by cells during respiration resulting in reduction of oxygen in exhaled air to 16%
- The remaining 79% of the air consist mainly of nitrogen, the percentage composition of which does not change significantly during breathing.
- Inspiration air contains 0.04% carbon dioxide. Cells of the body produce carbon dioxide as a waste product during respiration. Expired air contains 4% carbon dioxide

Respiratory gases

- Respiratory gases are gases in air that are involved in respiration. These include;
 - ✓ Oxygen
 - ✓ Carbon dioxide
 - ✓ Water vapour

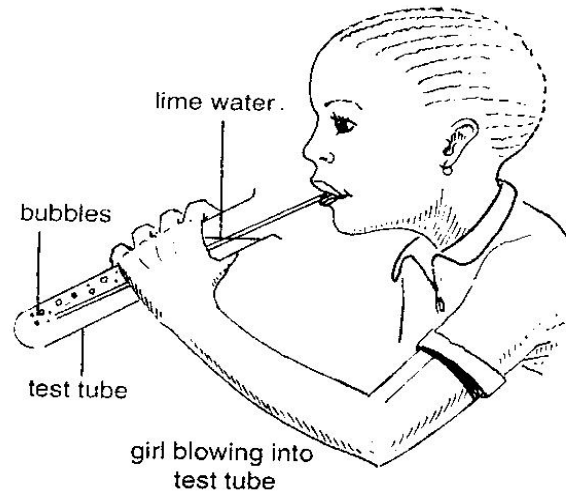
Experiment to compare oxygen in inhaled and exhaled air



- Place a candle inside a container with inhaled and exhaled air and measure the water displacement.
- Water displacement in inhaled air is greater and the candle burns for a longer time than in exhaled air. This is because inhaled air contains more oxygen and less carbon dioxide when compared to exhaled air which contains less oxygen and more carbon dioxide.
- When candles were burning they used up oxygen. Water rise is highest in jar with most amount of oxygen

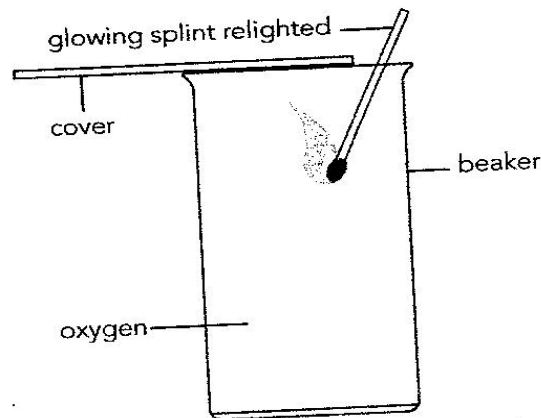
The test for carbon dioxide

- Blow air into limewater or bicarbonate indicator.
- Limewater changes from clear to milky and bicarbonate indicator changes from red to yellow.



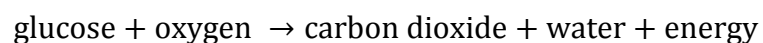
The test for oxygen

- Insert a glowing wooden splint into a test tube with oxygen.
- The glowing splint rekindles or relights in the presence of oxygen.



RESPIRATION

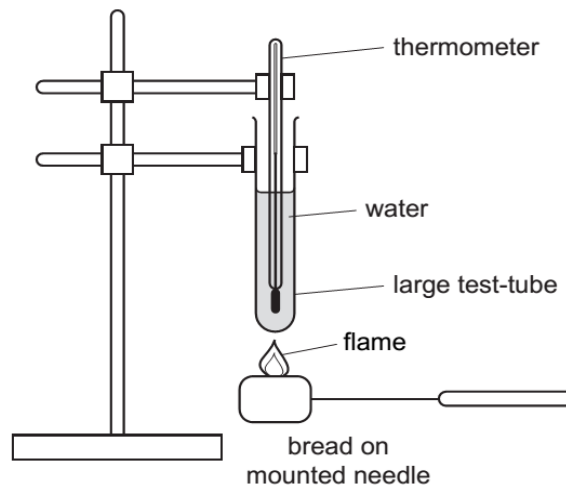
- It is the breakdown of carbohydrates to release energy



Experiment to show that energy is released during burning of during respiration

- Place some mealie-meal in a lid or crucible and heat the meal until it begins to burn.
- Light the meal in the dish and remove the burner so that the burn on its own.

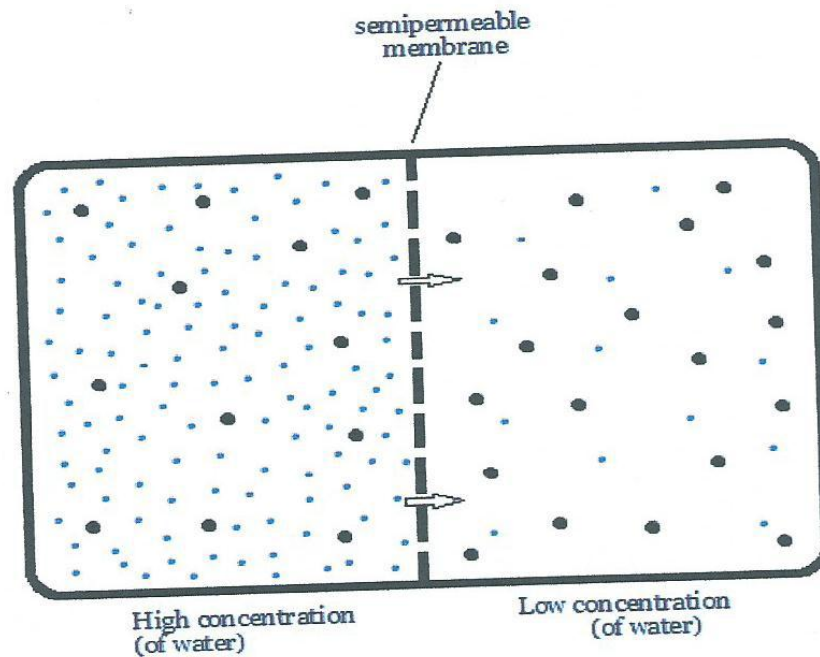
- Place a small container of water above the burning meal
- Record the temperature of the water before and after the meal has burnt away.
- The temperature increase is caused by the heat given off by the burning mealie meal. The meal had chemical energy stored in it.
- The energy trapped in plant material by photosynthesis is released in the body by respiration.



TRANSPORT SYSTEM

TRANSPORT IN PLANTS

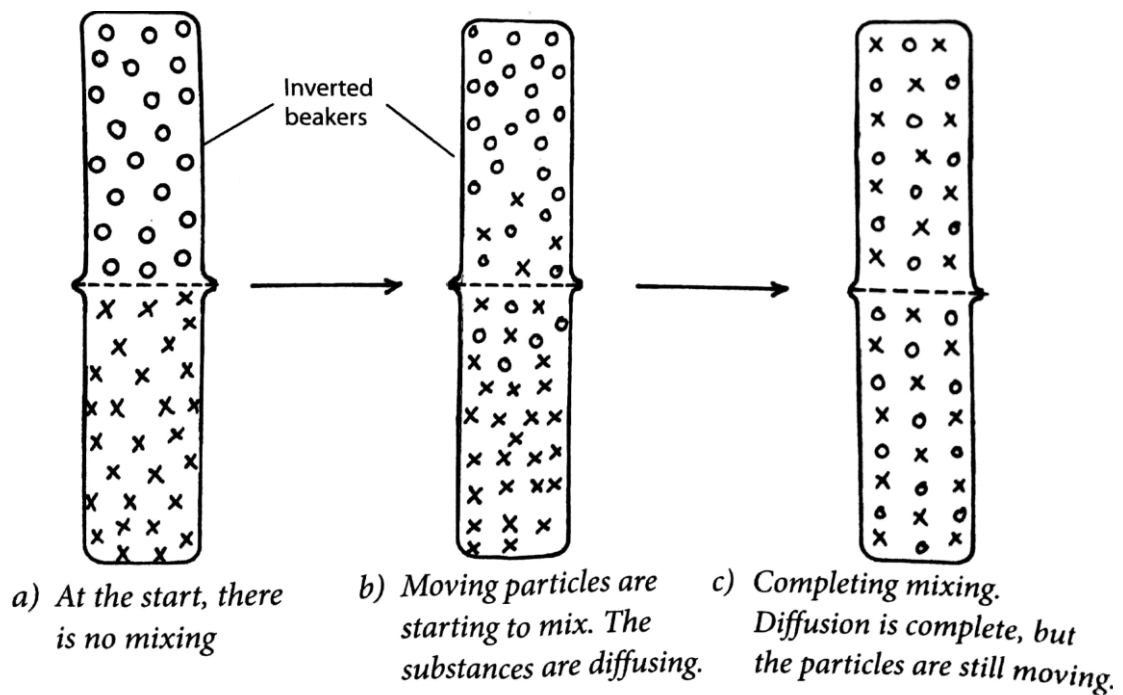
Osmosis



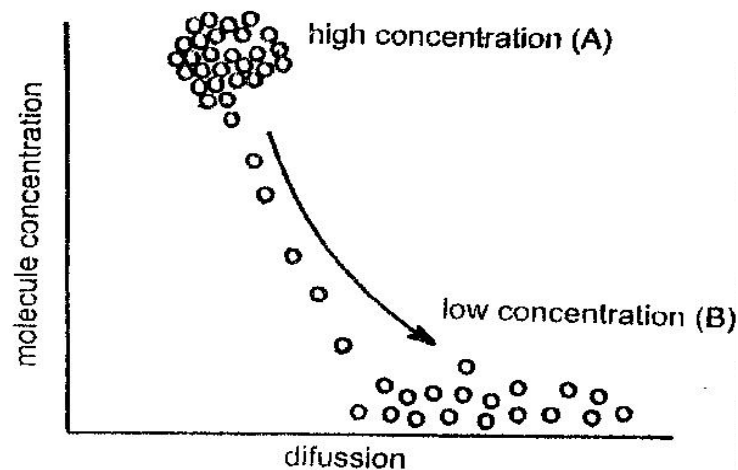
- Osmosis is the net movement of water molecules through a partially (semi) permeable membrane from an area of higher water concentration.
- A partially permeable membrane allows small molecules (water) to pass through them and prevents the larger molecules from passing through until the water particles are evenly spread out i.e. at equilibrium.

Diffusion

- Diffusion is the net movement of particles from a region of their concentration to a region of their lower concentration down a concentration gradient, as a result of their random movement.

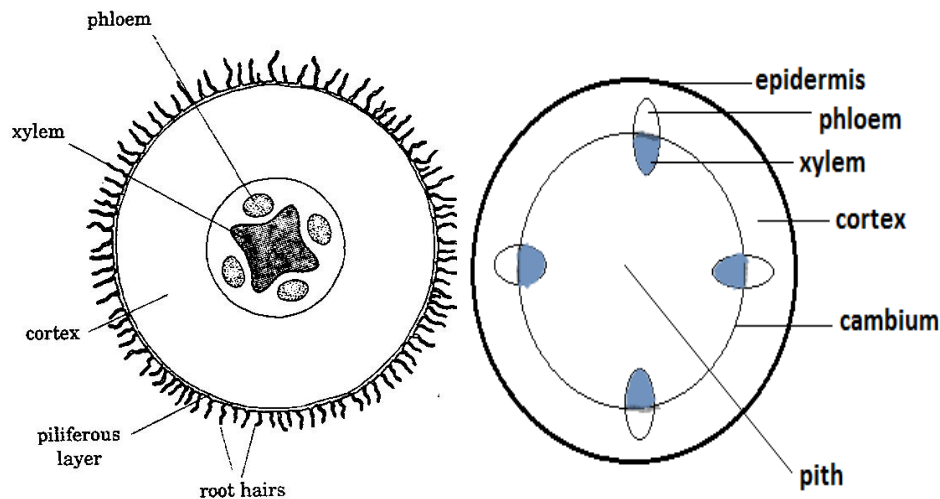


- Diffusion will always proceed whenever there are differences in concentration until the molecules are evenly distributed throughout the system
- The bigger the difference in concentration of a substance, the faster it will diffuse
- Molecules are in random motion but there is a net movement of molecules away from the more concentrated area.



ROOT AND STEM STRUCTURE

Internal structures of a dicotyledonous root and stem



Epidermis For absorption i.e. the cells have large central vacuoles and large surface areas for the uptake of water through osmosis and protect the inner root tissue.

Cortex Provide a pathway for water to the xylem, but can also store water and food.

Phloem Conduction of food materials (translocation).

Xylem Conduction of water and dissolved salts and support.

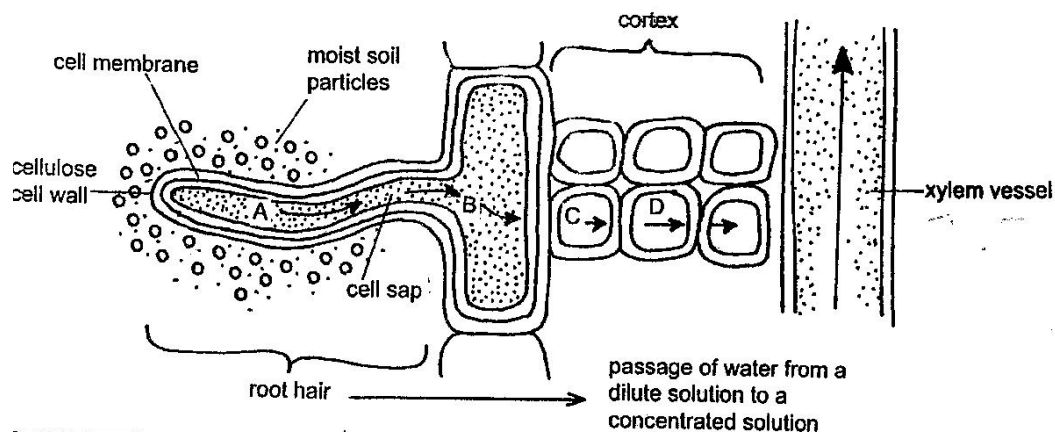
Cambium Formation of secondary xylem and phloem resulting in growth in diameter.

Root hairs they provide the main absorbing region for water and mineral salts from the soil. They are numerous to provide more surface area for absorption.

Water and ion uptake

- Water is drawn into the root hairs mainly by osmosis.
- The cell sap in the root hair cells is more concentrated with sugars and salts than the soil water. Water molecules are drawn across the permeable cellulose cell wall and semi-permeable membrane and protoplasm then into the vacuole.
- After water has been absorbed into the root, there are more water molecules in region A than B, so water diffuse from A to B then C and D until it reaches the xylem.
- Molecules or ions move from a region of low concentration to higher concentration i.e. they move against concentration gradient and energy is used to move molecules from the soil water into the root hair cells. This kind of movement of molecules is called active transport.

- The use of energy enables ions to move against the concentration gradient.

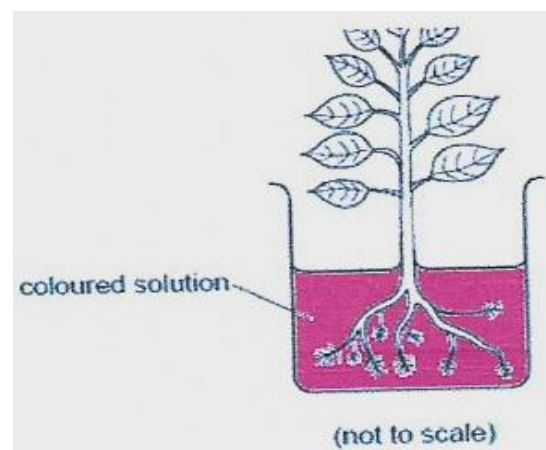
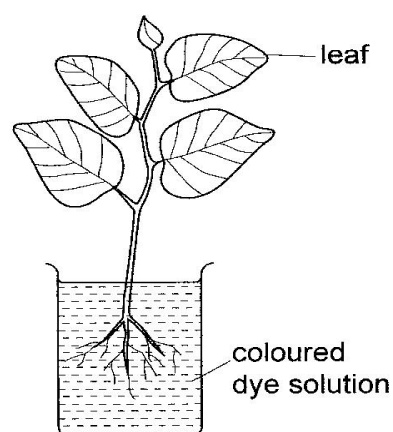


Water movement in plants

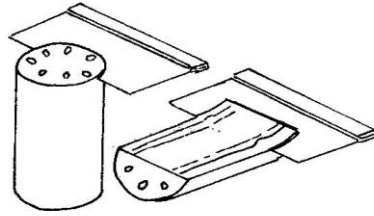
- The xylem vessels conduct water to the leaves where it diffuses to all cells.
- Water vapour evaporating from a leaf creates a kind of suction drawing a stream of water upwards, as water molecules are attracted to each other. This creates a transpiration stream, pulling water up from the root.
- Xylem vessels act like fine tubes drawing water up the stem by capillary action. Water molecules have high surface tension and hold to each other well enough to pull each other up.
- Roots also produce a root pressure i.e. the water absorbed osmotically by the roots, force water already present in the xylem upwards in the stem.

Demonstrating water movement in a plant

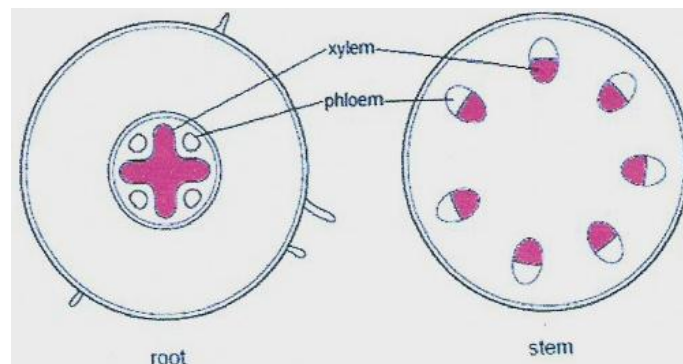
- To track the movement of water in a plant, a coloured solution can be used



- A young dicotyledonous plant was uprooted, placed in a beaker with coloured water, left in sunny place for a few hours
- Cut off thin slices of plant roots and stem to trace the movement of the dye.



- Examine these slices using a hand lens
- Make simple diagrams to show the distribution of the dye. The vascular bundles (xylem) will have been stained by the colouring since water moved by osmosis up the stalk.



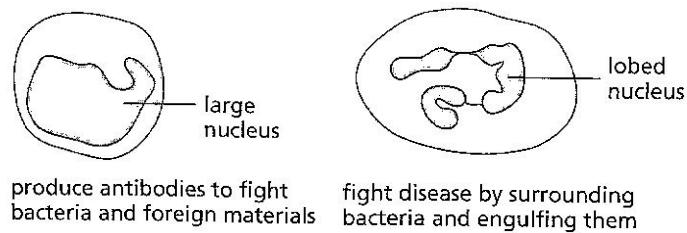
- The dye stain distribution pattern shows the pattern of xylem and hence vascular bundle distribution in the dicotyledonous stem and root.
- Water is drawn up in the plant through the xylem vessels which are part of the vascular bundle.

TRANSPORT IN HUMANS

CIRCULATORY SYSTEM

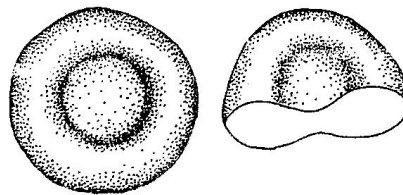
Blood components

White blood cells (leucocytes)



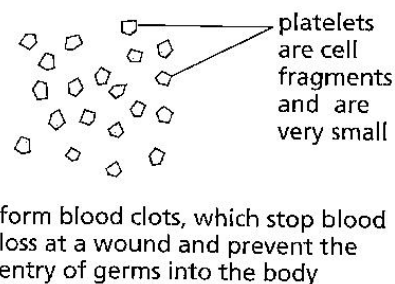
- Destroy bacteria and fight infection through phagocytosis and production of antibodies. They protect the body from an infection.
- Phagocytes engulf invading micro-organism by a process called phagocytosis i.e. they ingest and destroy pathogens to prevent infection.
- Lymphocytes produce antitoxins (antibodies) which neutralize the toxins produced by bacteria or agglutinate pathogens

Red blood cells (erythrocytes)



- Are small disc-like cell which don't have a nucleus. They contain hemoglobin, red pigment which combines with oxygen and is taken to all body tissues thus red blood cells transport oxygen.

Platelets (thrombocytes)



- Platelets are small structures which assist in blood clotting.
- Whenever a blood vessel is damaged, platelets secrete an enzyme (thrombokinase) which act on a blood protein (prothrombin) and turns it into thrombin. This in turn assist in the conversion of soluble fibrinogen in to insoluble fibrin which

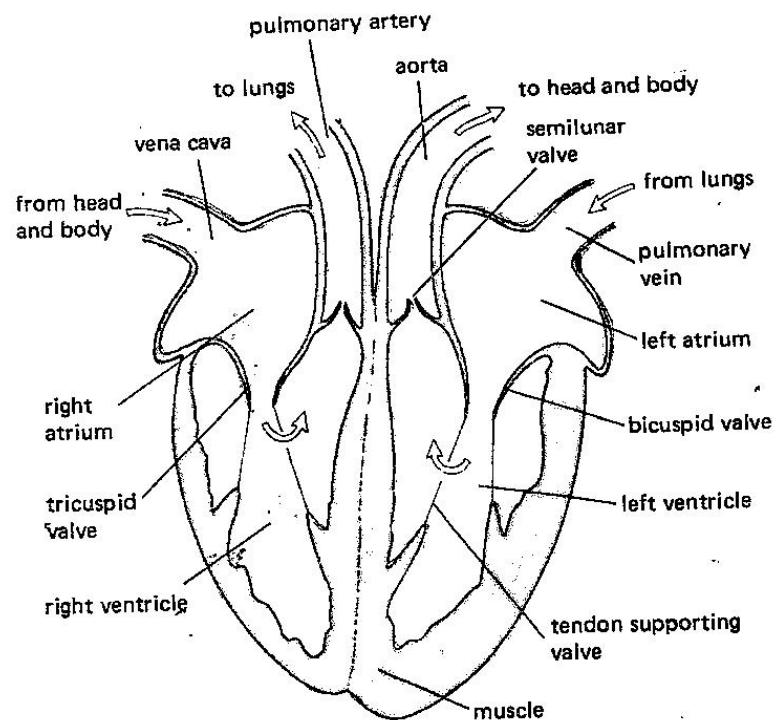
forms an obstruction in which the blood cells become entangled thus stopping breeding.

- The clot helps to prevent disease causing organisms from entering the body.

Plasma

- Plasma is a yellowish and slightly alkaline liquid part of the blood which carries the majority of the transport function.
- Plasma contains dissolved substance like blood proteins (fibrinogen), inorganic salts of (Ca, Na & K), organic constituents (glucose & urea) dissolved gases (oxygen & carbon dioxide) and secretions (hormones and enzymes)
- Blood plasma is a medium in which blood cells and platelets move to all parts of the body.

The heart



Functions of the heart

- The heart pumps and receives blood to and from all body parts.
- The heart is a muscular organ made of cardiac muscles that contract rhythmically to enable the heart to pump blood out.

- It consists of two atria – receiving chambers and two ventricles – pumping chambers.
- The septum separates the left and right sides preventing mixing of oxygenated and deoxygenated blood.
- Walls of the left ventricles are three times thicker than the right ventricle to generate enough pressure to move the blood to all organs of the body. Right ventricles only pumps blood to the lungs.
- Atrium opens into corresponding ventricles. Valves allow blood to flow in one direction or prevent back flow of blood.

Blood circulation

- Blood in the right ventricle is pumped to the lungs. Blood from the lungs flows back into the left atrium through the pulmonary vein and then is pumped into the left ventricle
- Blood in the left ventricle is pumped to the rest of the body through the aorta. Blood returns to the heart where it enters the right atrium through the vena cava. The right atrium pumps blood to the right ventricle.

The main blood vessels to and from the heart

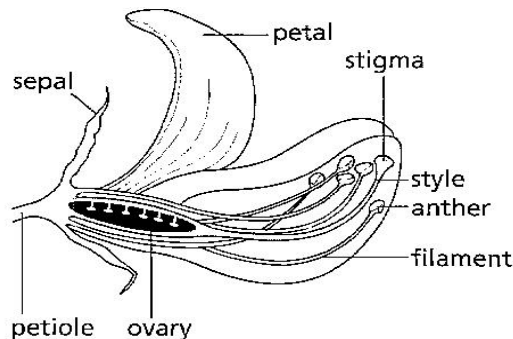
- Vena cava – carries deoxygenated blood to the heart
- Aorta – carry deoxygenated blood from the heart to the body
- Pulmonary artery – carry deoxygenated blood from the heart and lungs
- Pulmonary vein – carry oxygenated blood from lungs to the heart

REPRODUCTION

REPRODUCTION IN PLANTS

The structure of a flower

- A flower is the reproductive organ of plants. It consists of carpel (female parts) and the stamens (male parts).

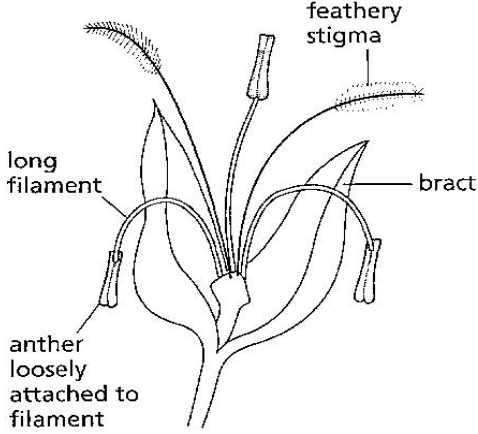
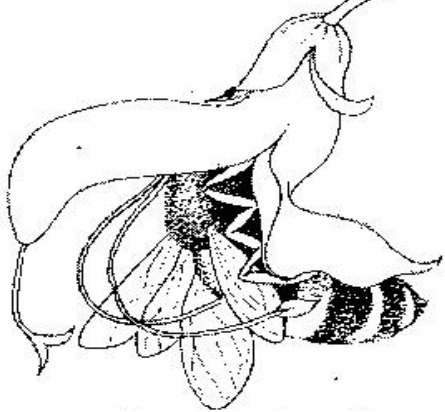


1. *Stamens* - consist of the anther and the filament. The anther produces pollen grains which carry the male sex cells. The filament holds and supports the anthers in position
2. *Carpel/pistils* - consist of a sticky top called stigma where pollen grains are deposited during pollination, a slender stalk (style) which supports the stigma and is the pathway for pollen tubes to ovules, and a swollen base (ovary) which contains ovules which carry the female sex cells.
3. *Filament* - hold and support the flower
4. *Sepal* - protects the flower when it is in bud
5. *Petal* - are often, but not always, large, scented and coloured.
6. *Receptacle* - supports the floral whorls

Pollination

- Pollination is the physical transfer of pollen grains from the anther to the stigma of the same species.
- It is important because it carries the male sex cells in the pollen grains to the female part of the flower
- There are two types of pollination i.e.
 1. *Self pollination* - takes place when pollen grains are transferred from the anther to the stigma of flowers within the same plant.
 2. *Cross pollination* - takes place when pollen grains are transferred to flowers from different plants but of the same species.

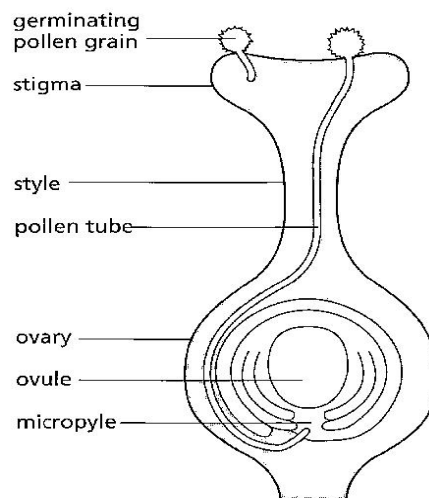
- Pollination is aided by insects or wind or birds

Wind pollinated flower	Insect pollinated flowers
 The diagram shows a cross-section of a wind-pollinated flower. It features a long, thin filament extending from the base. At the top of the filament is a feathery stigma. A bract is shown on the side. The anther is loosely attached to the filament. Labels include: 'long filament', 'feathery stigma', 'bract', and 'anther loosely attached to filament'.	 The diagram shows a cross-section of an insect-pollinated flower. A bee is shown inside the flower, reaching for nectar. The stigma and anther are visible. Labels include: 'stigma' and 'anther'.
- Abundant, dry or light pollen grains that can be carried by the wind - Petals are small to expose stamens and carpel to the wind. - Petals are dull (green) - Anthers and stigmas are long so they are exposed to the wind. - Stamens are exposed to the wind so that the pollen can be carried away by the wind. - Stigma feathery to provide large surface to trap as much pollen as possible from the wind.	- Have large coloured and scented petals to attract insects and birds. - Nectary produces sweet nectar which attracts insects. - Stigmas and anthers are enclosed inside the flower protected by petals. - Anthers have sticky pollen grains that stick to the insect's body. - Stigma is also sticky so that pollen grains on the insect's body stick to the stigma.

Fertilisation

- Fertilisation is the fusion of male and female sex cells to form a zygote.
- When pollen grain is deposited on the stigma, it germinates a pollen tube which grows down the style carrying the male nucleus along with it.

- Upon reaching the ovule, the tube breaks open and the male nucleus fuses with the female nucleus to form a zygote
- After fertilisation the following changes occur;
 1. Fertilised ovule divides by mitosis to form a seed containing the embryo plant (plumule & radicle) and food stores (cotyledons or endosperms)
 2. The wall of the ovule forms the seed testa (coat). The ovary walls develop into a fruit, which may be fleshy or dry pod.
 3. All other parts shrivel and die

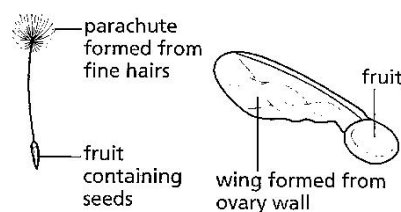


SEED DISPERSAL

- Seed dispersal is the scattering of seeds far away from the plant in order for new plants to stand a good chance of growing to maturity.

Wind dispersal

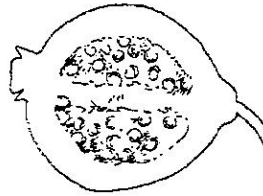
- Fruits usually have a wing or hairs to help them be carried by the wind from the parent plant e.g. sycamore and dandelion



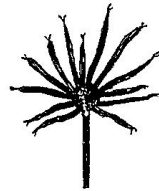
Animal dispersal

- There are two main modifications of fruits for animal dispersal i.e.

- Succulent fruits which attract animals because they have juicy flesh which is pleasant to eat but the seed is not palatable, so it is deposited in animals' droppings some distance from the parent plant e.g. tomatoes, guavas

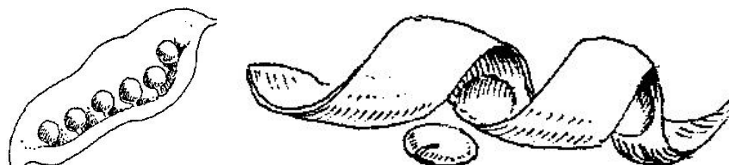


- Hooked fruits which catch or cling on to an animal's fur as it brushes past the parent plant. Eventually the seeds drop off, a further distance from the parent plant e.g. black jack



Mechanical (self) dispersal

- Pods often twist as they ripe because of changes in the amount of moisture in the air causing strains on the pod. The pod burst open suddenly and with enough force to throw its seeds some distance from the parent plant e.g. beans, peas

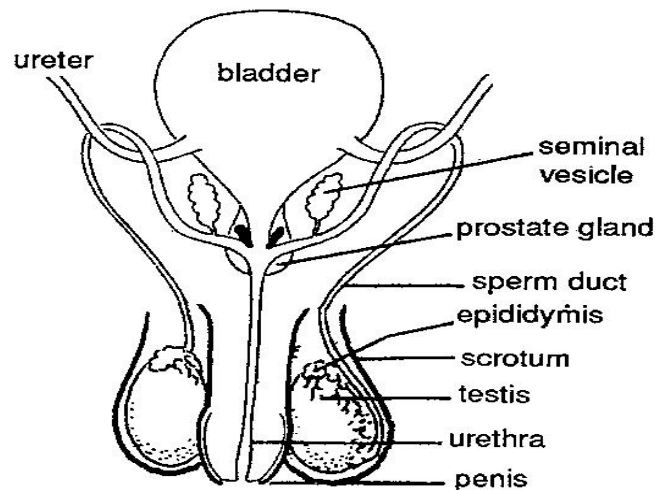


Water

- Fruits are light such that they float away and may eventually be washed on riversides where the seed develops into a new tree e.g. coconut.

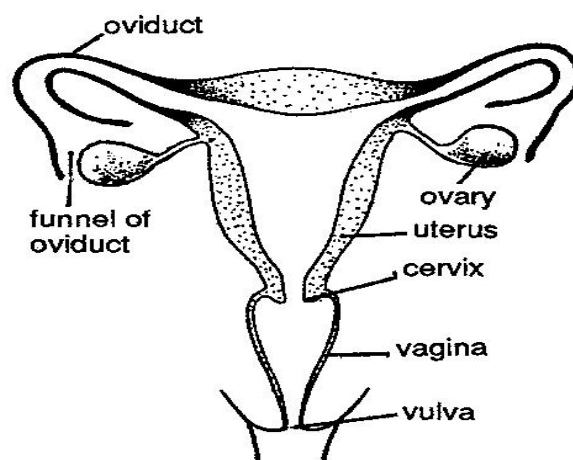
REPRODUCTION IN HUMANS

Male reproductive system



- Testes** Is the male gonad that makes and produce sperm cells. Produce male hormones which cause secondary sexual characteristics in males.
- Scrotum** It is a sac that holds testes outside the body keeping them cooler than body temperature to favour sperm production.
- Epididymis** Mass of tubes in which sperms are temporarily stored.
- Sperm duct** Is a muscular tube that links the testes to the urethra to allow the passage of semen containing sperms
- Urethra** Serves as a duct for passage of semen and urine through the penis.
- Penis** Consist of erectile and connective tissue with numerous blood vessels. During excitation, it becomes engorged with blood, stiffens and become erect, inserted into the vagina in order to transfer sperms.

Female reproductive system

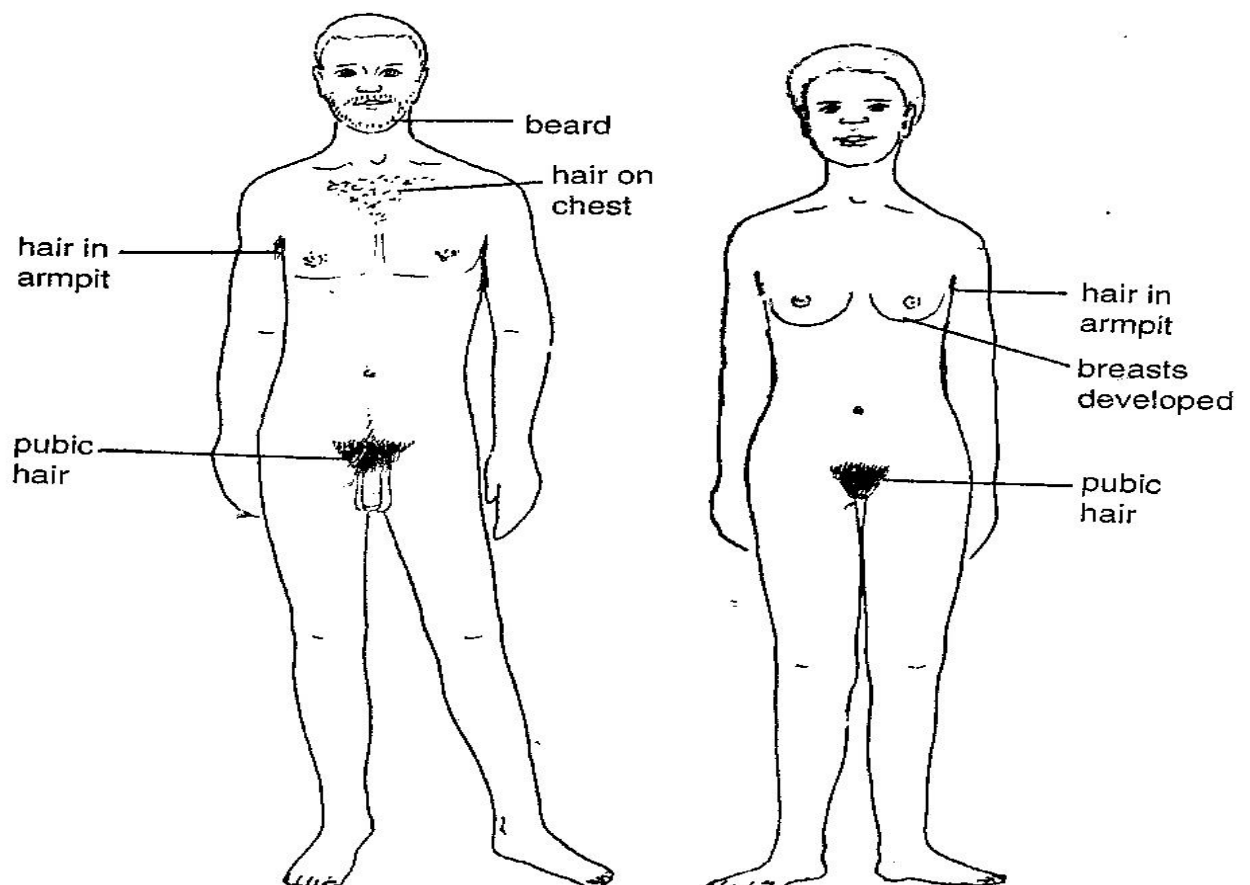


- Ovaries** Contain follicles in which ova or eggs are produced. They also produce the female hormones which causes secondary sexual characteristics in females

<i>Oviduct</i>	carries an ovum to the uterus, with propulsion provided by tiny cilia in the walls. This is also where fertilisation takes place.
<i>uterus</i>	Is where the fertilised ovum develops into a foetus during pregnancy
<i>cervix</i>	Is a narrow ring of muscle that closes the uterus but can expand greatly during birth of a baby.
<i>vagina</i>	Is a muscular organ which opens to the outside of the body. The vagina receive male penis during copulation and it is also where sperms are deposited during copulation.

PUBERTY

- Puberty is the stage at which sexual organs mature for reproduction.
- In males it starts between the age of 9 and 14 while in females it starts between 8-13 years



Signs of puberty

Boys

- Begins at around the age of 12 – 14 years. The signs include;
 - ✓ Body hairs grow e.g. in arm pits, on chest, face and pubic regions
 - ✓ Lengthening and widening of the penis
 - ✓ Increased mass and height
 - ✓ Deepening of voice due to enlarged voice box
 - ✓ Broadening of shoulders
 - ✓ Wet dreams occurs i.e. involuntary ejaculation of semen during sleep

Girls

- Begin at the age of about 10 to 13 years. Changes are mainly caused by the hormones oestrogen and progesterone. The signs include;
 - ✓ Onset of menstruation
 - ✓ Widening of the pelvic girdle (hips) to prepare the body for carrying and delivering a baby.
 - ✓ Enlarged breasts
 - ✓ Body hair grows i.e. in armpits and pubic regions
- Premenstrual symptoms include;
 - ✓ Mood swings and irritability or anger
 - ✓ Poor concentration
 - ✓ Social withdrawal
 - ✓ Change in libido
 - ✓ Tension and anxiety
 - ✓ Breast tender
 - ✓ Nausea and constipation or diarrhea
 - ✓ Period pain when the uterus contracts
 - ✓ Cramps
 - ✓ Acne

HEALTH AND DISEASES

HEALTH

- Health is complete state of physical, mental and social well being. Healthy is the well being of the body in its entirety i.e. with all the normal functions well coordinated.
- Mental well being is to do with a person's state of mind and social well being is to do with having decent accommodation, clothing, food and being part of a functional community and physical well being is to do with how well your body functions
- To enjoy good health, a person needs proper sleep and rest, proper shelter, balanced diet and enough exercises or physical fitness

DISEASES

- A disease can be defined as a disorder or malfunction of the body which leads to a departure from good health

Causes of disease

- *Poor nutrition* e.g. kwashiorkor
- *Pathogens* e.g. bilharzia (worms), malaria (protozoa), polio (virus) and typhoid (bacteria)
- *Chemicals and poison* e.g. lung cancer, asbestosis, liver cirrhosis, emphysema
- *Genetic defects* e.g. albinism, anaemia, colour blindness, Down's syndrome, dwarfism

Methods of transmission of diseases

- Disease transmission is the way in which a disease is spread. These include;

Contact

- Pathogens can be passed from one person to another when people are in close contact, directly or indirectly
- Direct requires close physical contact between an infected person and a susceptible person and the physical transfer of micro-organisms. This includes touching an infected person, kissing, and sexual contact, contact with oral secretions or body lesions. Diseases include Ebola, STIs etc

- Indirect contact transmission refers to situations where a susceptible person is infected from contact with contaminated surfaces e.g. clothes. Diseases include ring worm and Tinea, Ebola etc

Droplet infection

- Refer to situations where droplet nuclei or dust containing micro-organisms can remain suspended in air for long periods of time.
- Airborne transmission allows organisms to enter the upper and lower respiratory tracts
- Some diseases are transferred by infected droplets contacting surfaces of the eye, nose, or mouth.
- Droplets containing micro-organisms can be generated when an infected persons coughs or sneeze. E.g. TB, typhoid, influenza, polio, pneumonia, whooping cough

Vectors

- Are organisms capable of transmitting diseases e.g. mosquito, flies or ticks etc.
- vectors feed by biting through the skin and then sucking blood. If there were any pathogens in the saliva of the insect then these would be injected into the blood of the host animals.
- Micro-organisms could also be located on the outside surface of a vector or their faeces and spread through physical contact with food, a common touch surface, or a susceptible individual.
- Diseases include malaria, cholera

Contaminated water

- Water is a favorable medium for the dispersal of the organisms causing gastro-intestinal infections such as dysentery, cholera and typhoid.
- When the organisms reproduce in the gut, eggs are carried out with the faeces. If the faeces or urine carrying pathogens are deposited directly in water that is used for drinking, the organisms may thus infect large number of people.
- Disease organisms are washed into water supply and contaminate it.

Contaminated food

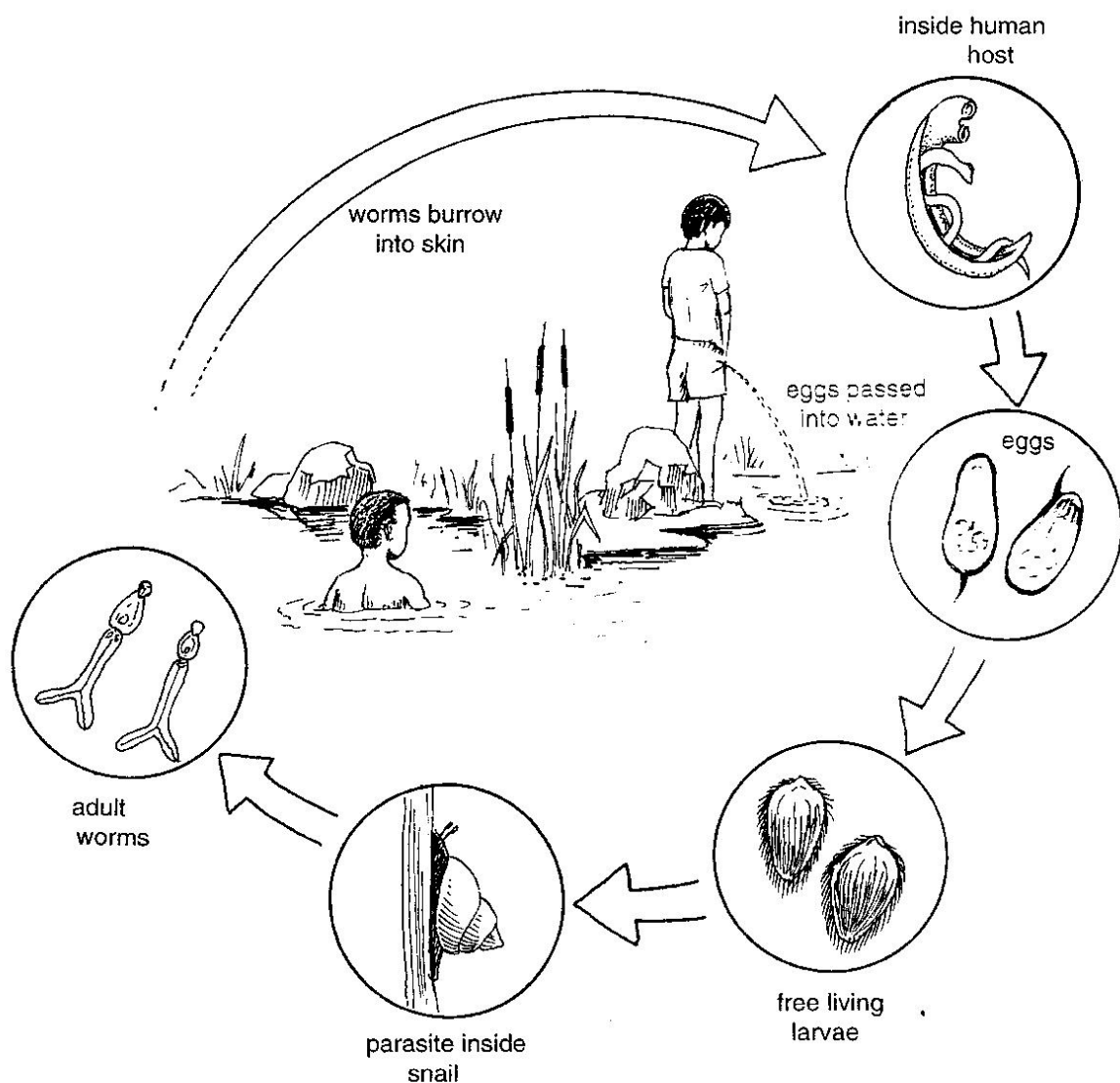
- Food may be contaminated with pathogenic organisms in a number of ways i.e. contaminated hands, flies and infested meat or washing in contaminated water.
- Diseases include typhoid, dysentery

BILHARZIA

Causes of Bilharzia

- Is caused by parasitic flat worm (blood fluke) called *Schistosoma* which invade blood vessels of the gut and the walls of the gut and walls of the urinary bladder.

Bilharzia Life Cycle



- The parasite's life cycle occurs in fresh water snail and a human

- Once flukes enter the body, they feed mainly on blood and tissue directly from the wall of the urinary bladder, to which they attach themselves by suckers.
- The flukes reproduce sexually and eggs are laid in the blood vessels of the gut or bladder.
- Each egg has a conspicuous spine and, as the embryo develops it vibrates, the spine cuts a hole in the wall of blood vessels. The eggs with a little blood enter the intestine or the bladder, to be dispersed either in faeces or in urine.
- The eggs will then hatch and develop into a series of different larvae inside the water snail. The larvae reproduce themselves asexually in the snail.
- The parasite leaves the snail and enters the water. It now has a forked tail to help it to swim.
- The larvae are able to penetrate a person who is bare footed in stagnant water or slow moving water bodies and immediately penetrate and invade the blood vessels. The larvae may also penetrate the body from the gut if they are swallowed in untreated drinking water.

Control or prevention

1. Avoid standing in water containing snails
2. Eliminate water snails by molluscides, which clear weed on which snails feed
3. Filtration and chlorination or boiling of water before use
4. Avoid urinating or defecating in water bodies
5. Use of drugs e.g. oxamiquine

HYGIENE

- Hygiene refers to conditions and practice that prevent the spread of disease.
- Good personal and food hygiene are important to prevent the spread of diseases.

Personal hygiene

- Involves taking care of your own body i.e.
 - Bathing daily using soap and water
 - Washing hands and scrubbing nails before preparation of food or eating
 - Clean the pubic and anal region

- Brushing teeth twice a day
- Washing and ironing clothes frequently
- Such simple precautions greatly reduce the chances of picking up an intestinal disease and also prevent passing on of diseases
- It also helps prevent fungal infections and will remove body parasites which might be carrying disease organisms.

Food hygiene

- Involves handling, serving, storing and preparation of food in a way that prevents the spread of diseases.
- It is important that a person who is working with food washes his/her hands with soap and water before preparing the food.
- Food should be stored at correct temperature and should be cooked correctly and thoroughly to kill bacteria. Food that has passed its expiry date should not be eaten

Toilet hygiene

- The toilet is a place where bacteria can be found in the toilet bowl, on the seat, under the lid, on the handle and on the floor.
- A firm bristled toilet brush is usually used to clean any waste and disinfectants are used to kill bacteria in the toilet.

WASTE DISPOSAL

- Litter is rubbish such as paper, tins and bottles that are left lying around.
- Wastes refer to any material that is unwanted i.e. domestic or industrial
- Some wastes can be naturally broken down by organisms and are called biodegradable waste. Some cannot be broken down naturally and are called non biodegradable wastes
- If wastes are not disposed of properly, it creates a health hazard. The wastes attract flies, mosquito and rats that spread diseases
- All wastes needs to be removed or disposed of. Methods of disposal include;

Burning

- Wastes are burnt in bins or incinerators

Advantages

- Reduces the volume of wastes
- Less expensive
- Prevents contamination of ground water by chemicals
- Pathogens destroyed
- Easy and takes less time (fast)

Disadvantages

- Causes air pollution
- Produce harmful chemicals
- Time consuming to separate wastes that can not be burnt
- Leads to greenhouse effect and hence global warming
- Expensive to control emissions into the air

Burying

- waste is buried in a managed landfill sites.

Advantages

- Cheapest form of solid waste disposal
- Filled land can be used for other purposes
- Can handle large amount of wastes
- Prevents bad odour that emanate from decomposing wastes
- Prevents insects and rodents from breeding in the wastes and cannot spread diseases

Disadvantages

- Requires large areas of land
- Contaminates the soil and or groundwater
- Poorly managed harmful substances create a serious health risk to humans and animals
- Emits greenhouse gases
- Can attract disease carrying animal

Recycling

- Recycling means to use waste materials to make other products. Waste paper, plastics, metal and glass can be recycled and remoulded into the same or other manufactured goods. Wastes for recycling are usually sorted in recycling bins.

Advantages

- Reduces the amount of wastes that needs to be disposed of.
- Conserves natural resources, since there is less demand for raw materials
- Creates income or employment
- Reduces pollution

Disadvantages

- Could be unhygienic
- Sorting can be time consuming and expensive